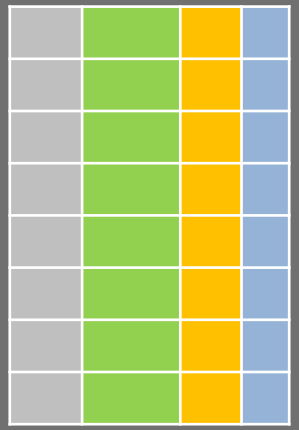
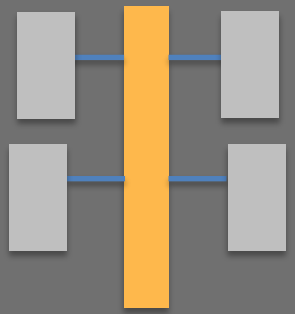
NoSQL Models

SQL Databases

Relational (OLTP)

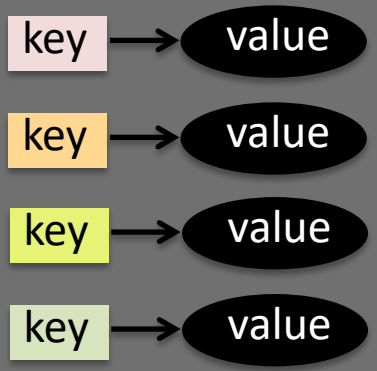


Analytical (OLAP)

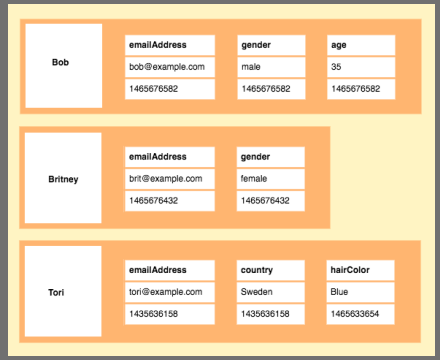


Non-SQL Databases

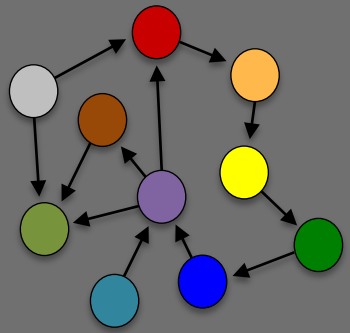
Key-Value



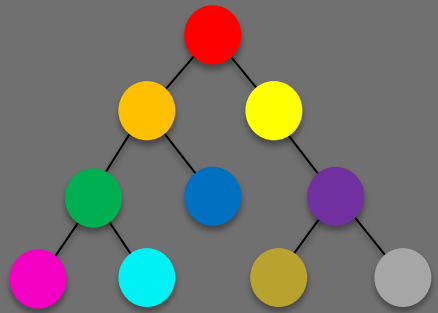
Column-Family



Graph



Document



Key-value store

Mapping

- Unique keys
- Mimic tables by constructing keys like “cust:1:lname”
- Mainly for fast lookup with known keys

Simple & fast









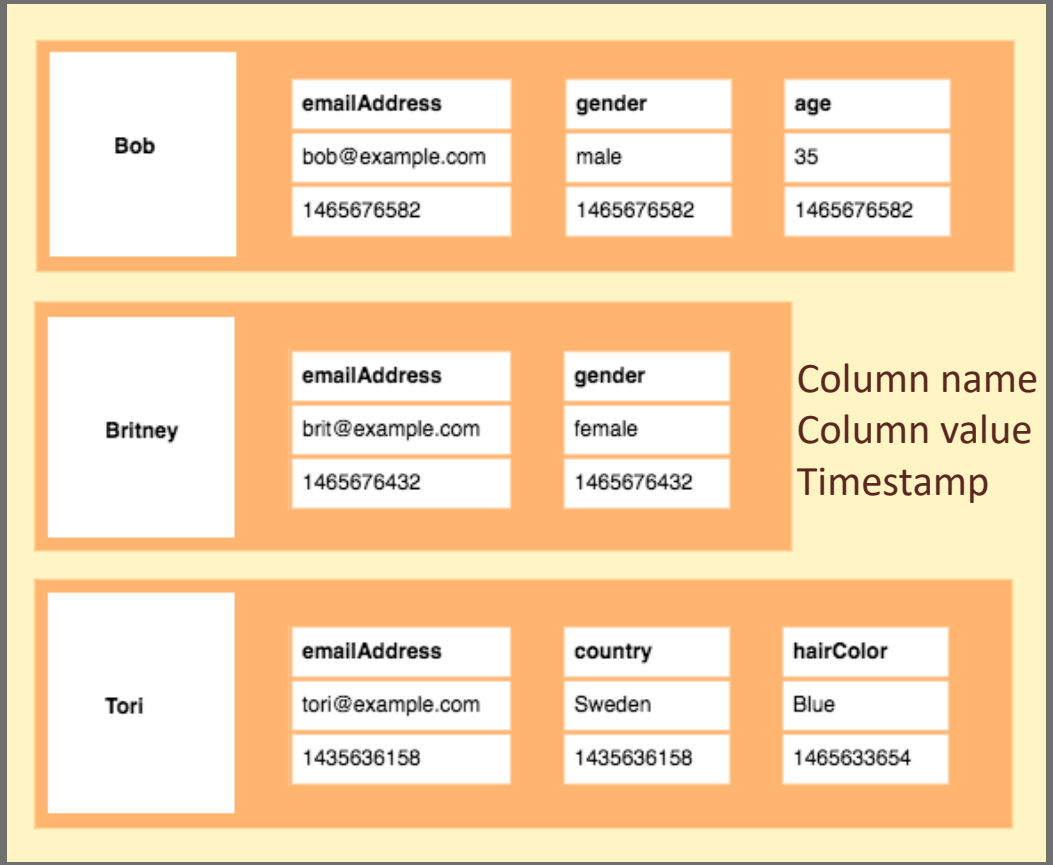






Column store

Column family
=
multiple rows



Column name
Column value
Timestamp

teradata.

Google BigTable

HYPERTABLE INC

cassandra

APACHE HBASE

Document

Collection of key-value pairs

- XML, JSON
- Analogous to a table row
- Hierarchical: Value can be embedded key-value pairs

Type most similar to relational, except …

- Heterogenous
- No many-to-many













Graph

Graph

- Nodes + edges
- Labeled with key-value pairs

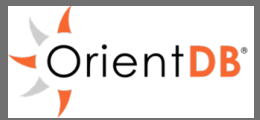
Some uses:

- ML / data mining
- Fraud detection
- Recommendations
- ID/access management
- Supply chain









Need multiple data models?

Use multiple DBMS.

Use a multi-modal DBMS.

Implementation

Vary, but…

Distributed

Horizontally scalable – Add more processors!

Often, relaxed consistency